

25) Show that $\neg(p \leftrightarrow q)$ and $\neg p \leftrightarrow q$ are logically equivalent.

Soln: Let $\neg(p \leftrightarrow q)$ be True. By defn. of negation, $\neg(\neg(p \leftrightarrow q))$ be False. By double-negation Law, $\neg(\neg(p \leftrightarrow q)) \equiv (p \leftrightarrow q) \therefore (p \leftrightarrow q)$ be False. By defn. of bi-conditional either p is True, q is False or p is False, q is True. Let $\neg p \leftrightarrow q$ be True. By defn. of bi-conditional, $\neg p, q$ is True or $\neg p, q$ is False. If $\neg p$ is True, by defn. of negation, $\neg(\neg p)$ is False. By double-negation law, $\neg(\neg p) \equiv p \therefore p$ is False. Similarly, we can show that if $\neg p$ is False, p is True. $\therefore \neg p \leftrightarrow q$ is True only if ① p is False, q is True OR ② p is True, q is False. $\therefore \neg(p \leftrightarrow q)$ and $\neg p \leftrightarrow q$ are logically equivalent $\square$

26) Show that $(p \rightarrow q) \wedge (p \rightarrow r)$ and $p \rightarrow (q \wedge r)$ are logically equivalent.

Soln: Let $(p \rightarrow q) \wedge (p \rightarrow r)$ be True. By defn. of AND, $(p \rightarrow q)$ is True and $(p \rightarrow r)$ is True. The following cases arise: ① Both q and r are True. In this case, no matter the value of p , $p \rightarrow q$ is True and $p \rightarrow r$ is True by defn. of conditional. ② q is True, r is False. Since $p \rightarrow r$ is True, by defn. of conditional, p is False. ③ q is False, r is True. Since $p \rightarrow q$ is True, by defn. of conditional, p is False. ④ Both q and r are False. From cases 2, 3 we know p is False.

Let $p \rightarrow (q \wedge r)$ be True. By defn. of conditional, two cases arises:

① $(q \wedge r)$ is True. In this case, the truth value of p doesn't matter. By defn. of AND, q is True and r is True. ② $(q \wedge r)$ is False. By defn. of conditional, p is False. From the defn. of AND, the following 3 cases arise: ① q is True, r is False ② q is False, r is True ③ Both q, r are False.

$\therefore (p \rightarrow q) \wedge (p \rightarrow r)$ and $p \rightarrow (q \wedge r)$ are logically equivalent. $\square$